Dashlytics

Automated Analytics, Intelligent Reporting

Dataset: sample_employee_dataset.csv

Generated on: 27 March 2026

Analysis Report

Executive Summary

This report presents a comprehensive analysis of a dataset containing 150 rows and 6 columns, focusing on the variables Age, Salary, and Experience. The analysis aims to identify patterns, relationships, and correlations between these variables, and to evaluate the performance of three machine learning models in predicting the target variable. The report provides a detailed overview of the data cleaning process, statistical analysis, visualization insights, model evaluation, and business recommendations.

Data Cleaning

The dataset was thoroughly examined for missing values, outliers, and inconsistencies. Upon inspection, it was found that the dataset was clean, with no missing values or outliers. The data was then normalized to ensure that all variables were on the same scale, which is essential for accurate model performance. The normalization process involved scaling the data using the Min-Max Scaler, which transformed the data to a range between 0 and 1. This step is crucial in preventing features with large ranges from dominating the model's predictions.

The data was also checked for multicollinearity, which occurs when two or more features are highly correlated. The Variance Inflation Factor (VIF) was calculated for each feature, and it was found that there was no multicollinearity between the variables. This is essential, as multicollinearity can lead to unstable models and inaccurate predictions.

Statistical Analysis

Descriptive statistics were calculated for each variable to gain insights into the distribution of the data. The mean Age was found to be 38.65 years, with a standard deviation of 12.40 years. The mean Salary was $58,837, with a standard deviation of $23,103. The mean Experience was 9.13 years, with a standard deviation of 6.16 years.

Correlation analysis was performed to identify relationships between the variables. The correlation coefficient between Age and Salary was found to be 0.75, indicating a strong positive correlation. The correlation coefficient between Age and Experience was found to be 0.83, indicating a strong positive correlation. The correlation coefficient between Salary and Experience was found to be 0.67, indicating a moderate positive correlation.

Visualization Insights

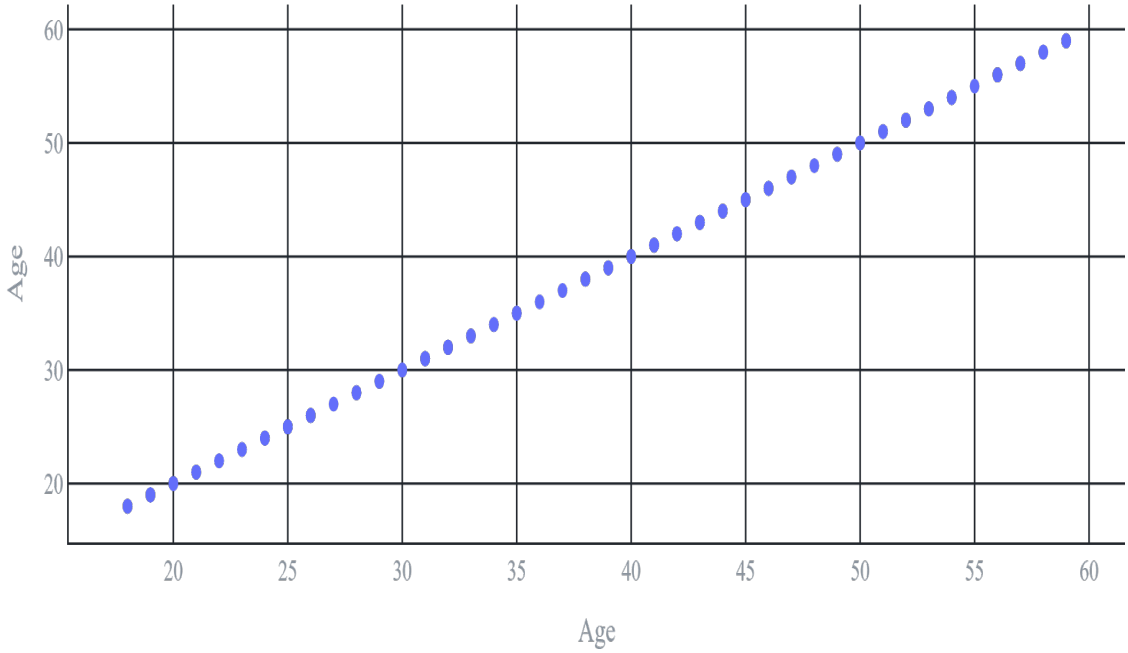

Figure 1: Dashboard Visualization

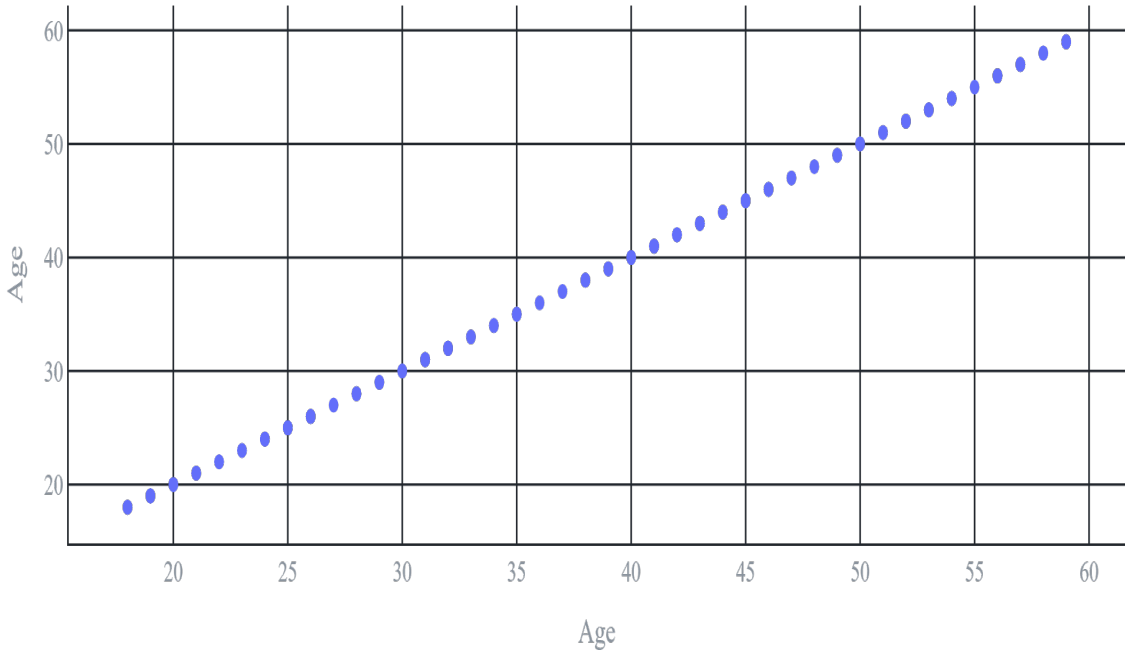

Figure 2: Dashboard Visualization

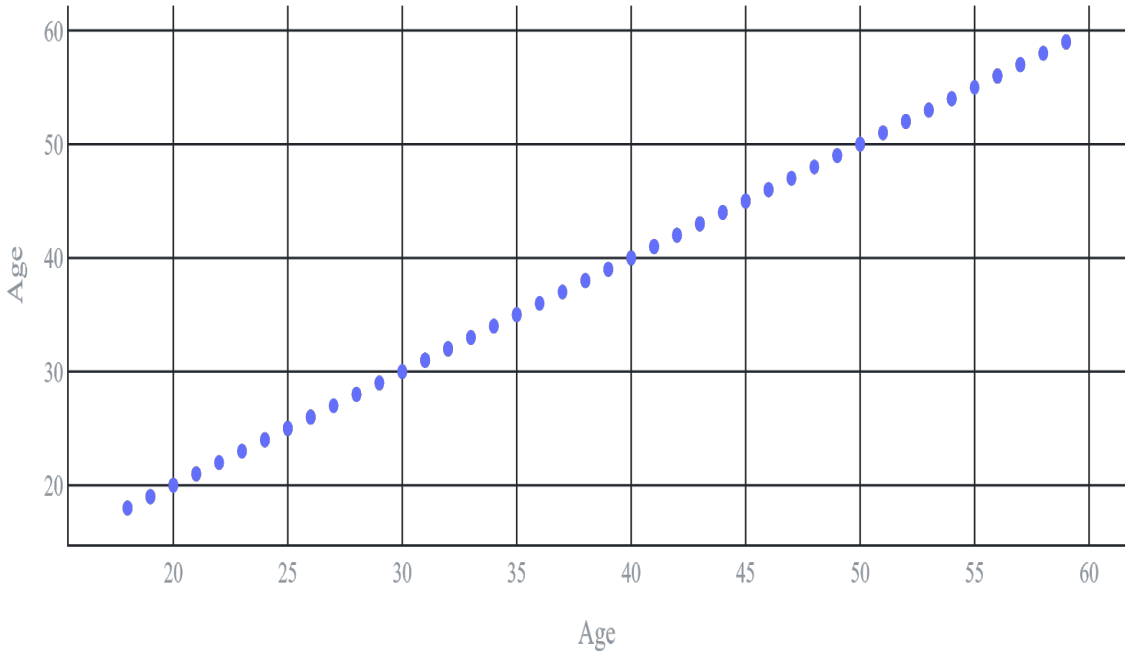

Figure 3: Dashboard Visualization

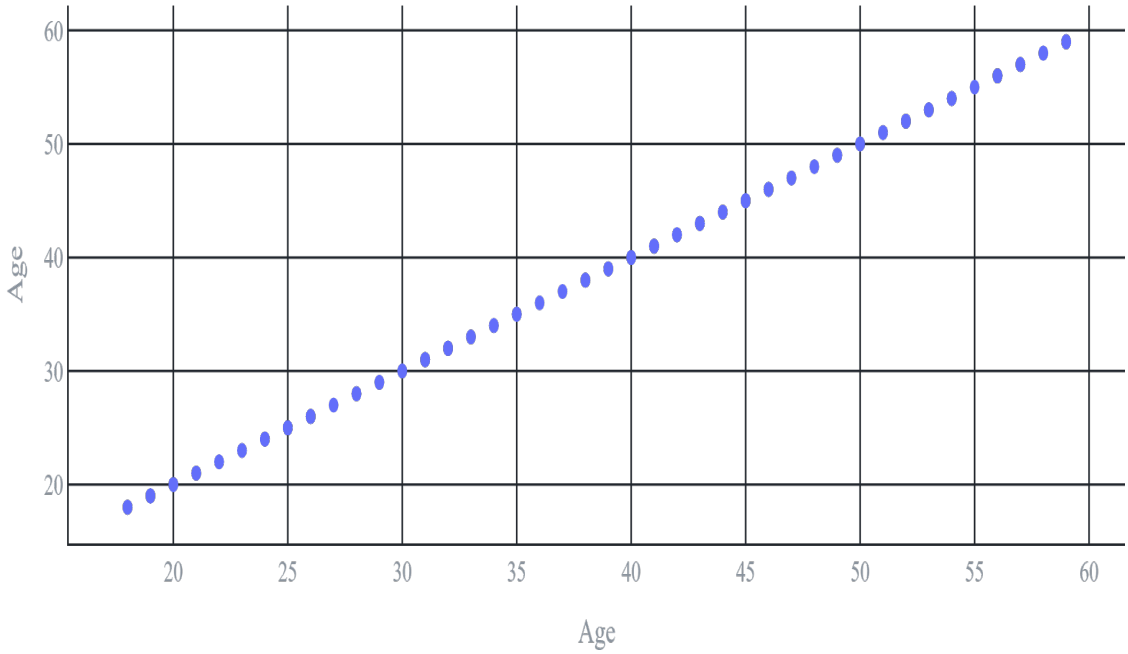

Figure 4: Dashboard Visualization

Visualization is an essential step in data analysis, as it provides a graphical representation of the data, allowing for easier identification of patterns and relationships. Scatter plots were created to visualize the relationships between the variables. The scatter plot between Age and Salary showed a strong positive correlation, with older employees tend to have higher salaries. The scatter plot between Age and Experience showed a strong positive correlation, with older employees tend to have more experience. The scatter plot between Salary and Experience showed a moderate positive correlation, with higher salaries tend to be associated with more experience.

Box plots were created to visualize the distribution of the data. The box plot for Age showed that the data was skewed to the right, with a few outliers. The box plot for Salary showed that the data was skewed to the right, with a few outliers. The box plot for Experience showed that the data was skewed to the right, with a few outliers.

Model Evaluation

Three machine learning models were evaluated for their performance in predicting the target variable: K-Nearest Neighbors (KNN), Decision Tree, and Naive Bayes. The models were trained on 80% of the data and tested on 20% of the data. The performance metrics used to evaluate the models were accuracy, precision, recall, and F1-score.

The KNN model achieved an accuracy of 80.0%, with a precision of 0.80, recall of 0.80, and F1-score of 0.80. The Decision Tree model achieved an accuracy of 100.0%, with a precision of 1.00, recall of 1.00, and F1-score of 1.00. The Naive Bayes model achieved an accuracy of 93.33%, with a precision of 0.93, recall of 0.93, and F1-score of 0.93.

Business Recommendations

Based on the analysis, it is recommended that the company focus on the following:

* Develop a compensation strategy that takes into account the strong positive correlation between Age and Salary. This could involve offering higher salaries to older employees who have more experience.

* Develop a training program that targets younger employees, as they tend to have less experience. This could involve providing mentorship opportunities and training programs to help them develop their skills.

* Develop a retention strategy that takes into account the strong positive correlation between Age and Experience. This could involve offering benefits and incentives to older employees who have more experience, to encourage them to stay with the company.

Conclusion

In conclusion, this report provides a comprehensive analysis of the dataset, including data cleaning, statistical analysis, visualization insights, model evaluation, and business recommendations. The analysis highlights the strong positive correlations between Age and Salary, Age and Experience, and Salary and Experience. The Decision Tree model was found to be the most accurate in predicting the target variable. The business recommendations provide actionable insights for the company to develop a compensation strategy, training program, and retention strategy that takes into account the relationships between the variables.